

```
!git clone https://github.com/hotuyen21pt/KLTN-Token-Merging.git
```

```
Cloning into 'KLTN-Token-Merging'...
remote: Enumerating objects: 217, done.
remote: Counting objects: 100% (217/217), done.
remote: Compressing objects: 100% (168/168), done.
remote: Total 217 (delta 105), reused 141 (delta 48), pack-reused 0 (from 0)
Receiving objects: 100% (217/217), 1.94 MiB | 5.71 MiB/s, done.
Resolving deltas: 100% (105/105), done.
```

```
!ls
```

```
sample_data  thesis_apc_baseline
```

```
!pip install -r thesis_apc_baseline/requirements.txt
```

```
Requirement already satisfied: srsly<3.0.0,>=2.5.3 in /usr/local/lib/python3.12/dist-packages (from spacy->pyabsa<3,>=2.4.
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3.12/dist-packages (from spacy->pyabsa<3,>=
Requirement already satisfied: weasel<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from spacy->pyabsa<3,>=2.4
Requirement already satisfied: confection<2.0.0,>=1.3.2 in /usr/local/lib/python3.12/dist-packages (from spacy->pyabsa<3,>
Requirement already satisfied: requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from spacy->pyabsa<3,>=
Requirement already satisfied: pydantic<3.0.0,>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from spacy->pyabsa<3,>=2
Requirement already satisfied: smmap<6,>=3.0.1 in /usr/local/lib/python3.12/dist-packages (from gitdb<5,>=4.0.1->gitpython
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingfac
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0-
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.6.3->metric
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.6.3->metric-vis
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.6.3->metri
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.6.3->metri
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.6.3->metric-visual
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.6.3->metric
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.0
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=13.8.0->typer-
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=13.8.0->type
Requirement already satisfied: blis<1.4.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.12->s
Requirement already satisfied: cloudpathlib>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from weasel<2.0.0,>=1.0.0->
Requirement already satisfied: smart-open>=5.2.1 in /usr/local/lib/python3.12/dist-packages (from weasel<2.0.0,>=1.0.0->sp
Requirement already satisfied: et-xmlfile in /usr/local/lib/python3.12/dist-packages (from openpyxl->metric-visualizer>=0.
Requirement already satisfied: webcolors in /usr/local/lib/python3.12/dist-packages (from tikzplotlib->metric-visualizer)=
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=13
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart-open>=5.2.1->weasel<2.0.0,>=1.
Downloading pyabsa-2.4.2-py3-none-any.whl (574 kB)
574.2/574.2 kB 20.9 MB/s eta 0:00:00
Downloading python_levenshtein-0.27.3-py3-none-any.whl (9.5 kB)
Downloading levenshtein-0.27.3-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (153 kB)
153.3/153.3 kB 19.5 MB/s eta 0:00:00
Downloading autocuda-0.16-py3-none-any.whl (5.1 kB)
Downloading boostaug-2.3.5-py3-none-any.whl (16 kB)
Downloading findfile-2.1.2-py3-none-any.whl (8.7 kB)
Downloading metric_visualizer-0.9.17-py3-none-any.whl (25 kB)
Downloading pytorch_warmup-0.2.0-py3-none-any.whl (11 kB)
Downloading update_checker-0.18.0-py3-none-any.whl (7.0 kB)
Downloading rapidfuzz-3.14.5-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (3.1 MB)
3.1/3.1 MB 94.8 MB/s eta 0:00:00
Downloading colorama-0.4.6-py2.py3-none-any.whl (25 kB)
Downloading tikzplotlib-0.10.1-py3-none-any.whl (54 kB)
54.2/54.2 kB 6.1 MB/s eta 0:00:00
Downloading xlswriter-3.2.9-py3-none-any.whl (175 kB)
175.3/175.3 kB 22.0 MB/s eta 0:00:00
Building wheels for collected packages: sequeval
  Building wheel for sequeval (setup.py) ... done
  Created wheel for sequeval: filename=sequeval-1.2.2-py3-none-any.whl size=16162 sha256=d5910032fc2fad2b1ea1ebbe28af127cb9d
  Stored in directory: /root/.cache/pip/wheels/5f/b8/73/0b2c1a76b701a677653dd79ece07cfabd7457989dbfbdc8d7
Successfully built sequeval
Installing collected packages: xlswriter, rapidfuzz, findfile, colorama, autocuda, update-checker, Levenshtein, tikzplotl
Successfully installed Levenshtein-0.27.3 autocuda-0.16 boostaug-2.3.5 colorama-0.4.6 findfile-2.1.2 metric-visualizer-0.9
```

```
!ls
```

```
sample_data  thesis_apc_baseline
```

```
!python thesis_apc_baseline/src/train.py
```

```
[epoch 3/20] train_loss=nan
[dev] P=0.7591 R=0.7591 F1=0.7591
Writing model shards: 100% 1/1 [00:08<00:00, 8.16s/it]
Saved best checkpoint to /content/thesis_apc_baseline/checkpoints/gas_t5_ate/best (dev F1=0.7591)
[epoch 4/20] train_loss=0.1220
[dev] P=0.7565 R=0.7690 F1=0.7627
Writing model shards: 100% 1/1 [00:12<00:00, 12.39s/it]
Saved best checkpoint to /content/thesis_apc_baseline/checkpoints/gas_t5_ate/best (dev F1=0.7627)
[epoch 5/20] train_loss=0.1046
[dev] P=0.7476 R=0.7723 F1=0.7597
[epoch 6/20] train_loss=nan
[dev] P=0.7179 R=0.7393 F1=0.7285
[epoch 7/20] train_loss=nan
[dev] P=0.7484 R=0.7657 F1=0.7569
[epoch 8/20] train_loss=nan
[dev] P=0.7261 R=0.7525 F1=0.7391
[epoch 9/20] train_loss=nan
[dev] P=0.7284 R=0.7525 F1=0.7403
[epoch 10/20] train_loss=nan
[dev] P=0.7556 R=0.7756 F1=0.7655
Writing model shards: 100% 1/1 [00:21<00:00, 21.21s/it]
Saved best checkpoint to /content/thesis_apc_baseline/checkpoints/gas_t5_ate/best (dev F1=0.7655)
[epoch 11/20] train_loss=nan
[dev] P=0.7404 R=0.7624 F1=0.7512
[epoch 12/20] train_loss=nan
[dev] P=0.7564 R=0.7789 F1=0.7675
Writing model shards: 100% 1/1 [00:17<00:00, 17.26s/it]
Saved best checkpoint to /content/thesis_apc_baseline/checkpoints/gas_t5_ate/best (dev F1=0.7675)
[epoch 13/20] train_loss=nan
[dev] P=0.7372 R=0.7591 F1=0.7480
[epoch 14/20] train_loss=nan
[dev] P=0.7340 R=0.7558 F1=0.7447
[epoch 15/20] train_loss=0.0068
[dev] P=0.7372 R=0.7591 F1=0.7480
[epoch 16/20] train_loss=nan
[dev] P=0.7389 R=0.7657 F1=0.7520
[epoch 17/20] train_loss=0.0040
[dev] P=0.7532 R=0.7756 F1=0.7642
[epoch 18/20] train_loss=nan
[dev] P=0.7476 R=0.7723 F1=0.7597
[epoch 19/20] train_loss=0.0036
[dev] P=0.7468 R=0.7690 F1=0.7577
[epoch 20/20] train_loss=nan
[dev] P=0.7444 R=0.7690 F1=0.7565
Writing model shards: 100% 1/1 [00:23<00:00, 23.51s/it]
Saved final checkpoint to /content/thesis_apc_baseline/checkpoints/gas_t5_ate/last
Loading weights: 100% 257/257 [00:00<00:00, 4379.79it/s]
[test] P=0.7844 R=0.8045 F1=0.7943

=== Training complete ===
Best dev F1      : 0.7675
Best checkpoint  : /content/thesis_apc_baseline/checkpoints/gas_t5_ate/best
Test P/R/F1     : 0.7844 / 0.8045 / 0.7943
Wall time (sec) : 2202.338
```

```
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM
import torch

MODEL_PATH = "/content/thesis_apc_baseline/checkpoints/gas_t5_ate/best"

# Load model và tokenizer
tokenizer = AutoTokenizer.from_pretrained(MODEL_PATH)
model = AutoModelForSeq2SeqLM.from_pretrained(MODEL_PATH)

# Inference
sentence = "the food was delicious but the staff was rude"

inputs = tokenizer(
    sentence,
    return_tensors="pt",
    truncation=True,
    max_length=128
)

with torch.no_grad():
    outputs = model.generate(
        **inputs,
        max_length=64,
        num_beams=4
    )

prediction = tokenizer.decode(
    outputs[0],
    skip_special_tokens=True
)
```

```
print("Sentence:")  
print(sentence)  
print("\nPrediction:")  
print(prediction)
```

Loading weights: 100%

257/257 [00:00<00:00, 4566.98it/s]

Sentence:

the food was delicious but the staff was rude

Prediction:

(food)